Yusuke Izawa, Assistant Professor, PyPy Contributor

me@yizawa.com
https://me.yizawa.tokyo
〒 191-0065. 6 Chome-6, Asahigaoka, Hino, Tokyo , Japan.

Education

2020-2023 **Ph.D. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *Supporting multi-scope and multi-level compilation in a meta-tracing just-in-time compiler.* (GPA 3.49)

2018-2020 **M.Sc. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *Stack Hybridization: A Mechanism for Bridging Two Compilation Strategies in a Meta Compiler Framework.*

2014-2018 **B.Sc. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *BacCaml: A Meta-JIT Compiler Based on Both Tracing and Method JIT Compilations.*

Employment History

2025.06 – 2025.09 Heinrich-Heine-Universität Düsseldorf, Guest Researcher.

2024.04 – now Tokyo Metropolitan University, Assistant Professor.

2023.04 – 2024.03 IBM Research – Tokyo, Research Scientist.

2023.04 – 2023.04 JSPS Research Fellow PD. (declined)

2021.08 – 2021.10 IBM Research – Tokyo, Research Internship (Paid).

2021.04 – 2023.03 JSPS Research Fellow DC2.

2020.11 – 2023.03 Tokyo Institute of Technology, Dept. of Math. and Comp., Research Assistant.

Selected Grants, Honours and Scholarships

2026 **Grant-in-Aid for Acceleration Fund for International Collaborative Research (International Collaborative Research Strengthening)**.

2025 **Tokyo Metropolitan University, Funds for Research Stay in Overseas**.

Graint-in-Aid for Early-Career Scientists. Research expenses are convered by KAKENHI.

2024 **Graint-in-Aid for Research Activity Start-up**. Research expenses are convered by KAKENHI.

2023 **Research Fellowship for Young Scientists (JSPS PD).** Fellowship from the Japan Society for the Promotion of Science (JSPS), covering living expenses. Research expenses covered by KAKENHI. (declined)

2021 **Research Fellowship for Young Scientists (JSPS DC2)**. Fellowship from the Japan Society for the Promotion of Science (JSPS), covering living expenses. Research expenses covered by KAKENHI.

2020 **JST Strategic Basic Research Programs ACT-X**. Research expenses covered by Japan Science and Technology Agency (JST).

2019 **2nd Place, Graduate Category, ACM Student Research Competition, *Association for Computing Machinery*. [*]**

Selected Publications

Journal

1. Yusuke Izawa, Hidehiko Masuhara, and Carl Friedrich Bolz-Tereick. "A Lightweight Method for Generating Multi-Tier JIT Compilation Virtual Machine in a Meta-Tracing Compiler Framework (Artifact)." In: *Dagstuhl Artifacts Series* 11.2 (2025), 16:1–16:4. ISSN: 2509-8195. DOI: 10.4230/DARTS.11.2.16. URL: https://drops.dagstuhl.de/entities/document/10.4230/DARTS.11.2.16.
2. Yusuke Izawa, Hidehiko Masuhara, Carl Friedrich Bolz-Tereick, and Youyou Cong. "Threaded Code Generation with a Meta-Tracing JIT Compiler." In: *Journal of Object Technology* (2022), 2:1–11. ISSN: 1660-1769. DOI: 10.5381/jot.2022.21.2.a1. arXiv: 2106.12496.

Conference Proceedings

1. Yusuke Izawa, Junichiro Kadomoto, and Hidetsugu Irie. "VisMorph: A Live Programming Environment for Shape-Adaptive Computers." In: *37th ACM Symposium on User Interfance and Software Technology (UIST)*. 2025.
2. Yusuke Izawa, Hidehiko Masuhara, and Carl Friedrich Bolz-Tereick. "A Lightweight Method for Generating Multi-Tier JIT Compilation Virtual Machine in a Meta-Tracing Compiler Framework." In: *39th European Conference on Object-Oriented Programming (ECOOP 2025)*. Ed. by Jonathan Aldrich and Alexandra Silva. Vol. 333. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 16:1–16:29. ISBN: 978-3-95977-373-7. DOI: 10.4230/LIPIcs.ECOOP.2025.16. arXiv: http://arxiv.org/abs/2504.17460. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ECOOP.2025.16.
3. Yusuke Izawa and Hidehiko Masuhara. "Amalgamating Different JIT Compilations in a Meta-Tracing JIT Compiler Framework." In: *Proceedings of the 16th ACM SIGPLAN International Symposium on Dynamic Languages*. DLS 2020. Virtual, USA: Association for Computing Machinery, Nov. 17, 2020, pp. 1–15. ISBN: 9781450381758. DOI: 10.1145/3426422.3426977.

Selected Academic Services

2026	Program Committee, PEPM 2026.
2025 – now	Steering Committee, VMIL 2025.
2025	Organizer and Program Co-Chair, VMIL 2025.
	Program Committee, Programming and Programming Language Workshop (PPL).
	Programming and Programming Language Workshop (PPL), Program Committee.
	Program Committee, PEPM 2026.
	Member, Steering Committee, VMIL 2025.
	Program Co-Chair, VMIL 2025.
	Programming and Programming Language Workshop (PPL), Program Committee.
2024	External Reviewer, The Programming Journal, Volume 8. Issue 2.
	Reviewer, ACM Transactions on Architecture and Code Optimization.
	Reviewer, ACM Transactions on Software Engineering and Methodology.
	Program Committee, MPLR 2024, ICCQ 2024, MoreVMs 2024.
2023	Program Committee, ICSME 2023 (Industry Track), ICCQ 2023.
	Artifact Evaluation Committee, The Programming Journal, Volume 8.
2022	Artifact Evaluation Committee, The Programming Journal, Volume 7.
2023	Artifact Evaluation Committee, The Programming Journal, Volume 8.

Teaching Experience (Lead and Co-Lead)

Experiment on Compiler Construction

- Target: BEng 2nd year
- Teaching a compiler construction by building a comiler from *While* language to WebAssembly in OCaml

Experiment on Compiler Construction

- Target: BEng 3nd year
- Teaching hardware proramming by using a FPGA board and Verilog

Research Seminar on Programming Language and Virtual Machine

- Target: BEng 3nd year
- Leading a seminar that discusses programming language implementation and virtual machine construction

Introduction to Programming in C (II) (Co-Lead)

- Target: BEng 2nd year
- Teaching programming in C

Excersixe on Scripting Languages (Co-Lead)

- Target: BEng 3rd year
- Teaching programming in Python and small interpreter construction